

## BLONDUOS, Iceland

WMO: 041450

Lat: 65.6580N

Lon: 20.2925W

Elev: 72

StdP: 14.66

Time Zone: 0.00 (E00)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>12     | (b)<br>10.9 | (c)<br>13.9 | (d)<br>4.0                    | (e)<br>6.8 | (f)<br>11.9 | (g)<br>7.3 | (h)<br>8.0 | (i)<br>15.3 | (j)<br>45.3           | (k)<br>35.4 | (l)<br>40.0 | (m)<br>35.5 | (n)<br>8.2            | (o)<br>90 | (p)<br>0.832 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.2                   | 62.0            | 54.3 | 59.7 | 53.0 | 57.7 | 51.8 | 55.7                | 60.7 | 54.0 | 58.5 | 52.6 | 56.8 | 12.0                 | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 52.9                            | 59.7 | 57.9 | 51.3 | 56.2 | 56.2 | 50.0 | 53.5 | 54.9 | 23.7          | 60.9 | 22.7 | 58.7 | 21.8 | 57.0 | 61.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 33.5              | 28.8 | 25.2 | 6.0                        | 67.4 | 3.5                | 3.0 | 3.5                                                | 69.5 | 1.5        | 71.2 | -0.4       | 72.9 | -2.9       | 75.1 |
|                   |      |      | 4.5                        | 58.5 | 3.6                | 1.6 | 2.0                                                | 59.7 | -0.1       | 60.6 | -2.1       | 61.5 | -4.7       | 62.6 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 38.9   | 31.2  | 30.9  | 31.8  | 36.3  | 41.9  | 47.4  | 50.5  | 49.1  | 45.8  | 38.3  | 32.9  | 29.7  |
|                                                                   | DBStd       | 9.69   | 6.97  | 7.67  | 6.25  | 6.14  | 5.59  | 4.54  | 3.97  | 4.35  | 5.71  | 6.63  | 6.90  | 7.90  |
|                                                                   | HDD50       | 4224   | 584   | 534   | 564   | 413   | 257   | 103   | 43    | 70    | 150   | 364   | 513   | 629   |
|                                                                   | HDD65       | 9540   | 1049  | 954   | 1029  | 862   | 717   | 527   | 450   | 493   | 575   | 827   | 963   | 1094  |
|                                                                   | CDD50       | 160    | 0     | 0     | 0     | 1     | 6     | 26    | 58    | 42    | 25    | 2     | 0     | 0     |
|                                                                   | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | WSAvg       | 11.6   | 13.1  | 13.3  | 12.6  | 11.4  | 10.8  | 10.2  | 9.1   | 9.6   | 10.9  | 11.5  | 12.3  | 14.1  |
|                                                                   | PrecAvg     | 21.7   | 2.0   | 1.9   | 1.9   | 1.1   | 1.2   | 1.4   | 1.6   | 2.0   | 2.3   | 2.2   | 2.0   | 2.0   |
| Precipitation                                                     | PrecMax     | 26.6   | 3.9   | 2.9   | 4.4   | 2.2   | 2.6   | 3.7   | 2.7   | 3.3   | 3.6   | 5.8   | 3.7   | 3.8   |
|                                                                   | PrecMin     | 17.9   | 0.7   | 0.7   | 0.6   | 0.1   | 0.2   | 0.2   | 0.6   | 0.5   | 1.0   | 0.9   | 0.4   | 0.4   |
|                                                                   | PrecStd     | 2.5    | 0.8   | 0.6   | 1.0   | 0.5   | 0.7   | 0.8   | 0.5   | 0.9   | 0.8   | 1.1   | 0.8   | 0.9   |
|                                                                   | PrecStd     | 2.5    | 0.8   | 0.6   | 1.0   | 0.5   | 0.7   | 0.8   | 0.5   | 0.9   | 0.8   | 1.1   | 0.8   | 0.9   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 50.0   | 48.3  | 48.5  | 56.1  | 60.0  | 62.7  | 65.9  | 64.2  | 62.9  | 56.5  | 53.1  | 49.1  | 49.1  |
|                                                                   | 0.4% MCWB   | 42.5   | 42.4  | 43.4  | 47.8  | 49.6  | 53.3  | 56.0  | 56.4  | 54.7  | 49.0  | 47.7  | 44.6  | 44.6  |
|                                                                   | 2% DB       | 46.3   | 45.9  | 45.6  | 50.8  | 55.7  | 59.1  | 62.1  | 61.0  | 59.2  | 53.7  | 47.4  | 45.9  | 45.9  |
|                                                                   | 2% MCWB     | 41.9   | 41.3  | 41.6  | 43.5  | 47.5  | 52.0  | 54.5  | 54.7  | 52.6  | 47.1  | 42.8  | 41.9  | 41.9  |
|                                                                   | 5% DB       | 43.6   | 44.1  | 43.1  | 48.0  | 53.2  | 56.8  | 59.6  | 58.4  | 56.9  | 51.2  | 44.9  | 43.5  | 43.5  |
|                                                                   | 5% MCWB     | 39.9   | 40.1  | 39.4  | 41.9  | 46.6  | 50.7  | 53.5  | 52.5  | 51.1  | 45.8  | 41.2  | 40.2  | 40.2  |
|                                                                   | 10% DB      | 40.6   | 41.9  | 40.7  | 45.5  | 50.7  | 54.7  | 57.2  | 56.2  | 54.5  | 48.6  | 42.3  | 40.6  | 40.6  |
|                                                                   | 10% MCWB    | 37.1   | 38.4  | 37.3  | 40.6  | 45.3  | 49.4  | 52.2  | 51.2  | 49.7  | 44.3  | 39.2  | 37.7  | 37.7  |
|                                                                   | 0.4% WB     | 43.6   | 44.1  | 44.1  | 45.9  | 50.8  | 55.2  | 57.7  | 57.6  | 55.9  | 50.2  | 48.1  | 45.6  | 45.6  |
|                                                                   | 0.4% MCDB   | 47.1   | 46.6  | 48.1  | 51.5  | 57.5  | 58.8  | 63.4  | 62.5  | 61.6  | 55.5  | 53.3  | 49.3  | 49.3  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 41.5   | 41.7  | 41.9  | 43.5  | 48.6  | 53.2  | 56.1  | 55.2  | 53.4  | 47.9  | 43.1  | 42.4  | 42.4  |
|                                                                   | 2% MCDB     | 45.4   | 45.3  | 45.4  | 47.8  | 54.2  | 57.5  | 61.4  | 60.0  | 58.2  | 52.7  | 46.3  | 45.6  | 45.6  |
|                                                                   | 5% WB       | 39.3   | 40.1  | 39.8  | 41.7  | 47.2  | 51.6  | 54.3  | 53.5  | 51.8  | 46.5  | 40.5  | 40.3  | 40.3  |
|                                                                   | 5% MCDB     | 42.7   | 43.8  | 42.8  | 45.8  | 52.5  | 56.2  | 58.8  | 57.9  | 56.4  | 51.1  | 43.1  | 43.4  | 43.4  |
|                                                                   | 10% WB      | 37.2   | 38.5  | 37.6  | 40.1  | 45.7  | 50.0  | 52.8  | 51.8  | 50.2  | 44.8  | 38.4  | 37.5  | 37.5  |
|                                                                   | 10% MCDB    | 40.1   | 41.8  | 40.6  | 44.2  | 50.5  | 54.0  | 56.6  | 55.6  | 54.6  | 48.6  | 41.1  | 40.2  | 40.2  |
| Mean Daily Temperature Range                                      | MDBR        | 10.1   | 10.1  | 10.4  | 11.3  | 10.9  | 10.5  | 10.2  | 10.2  | 10.4  | 9.9   | 9.3   | 10.5  | 10.5  |
|                                                                   | 5% DB       | 11.2   | 11.1  | 12.4  | 14.3  | 14.2  | 14.0  | 15.4  | 13.5  | 12.3  | 12.8  | 10.9  | 11.3  | 11.3  |
|                                                                   | 5% WB       | 8.8    | 9.1   | 9.8   | 9.6   | 8.7   | 8.8   | 9.5   | 8.5   | 8.4   | 9.6   | 9.2   | 9.7   | 9.7   |
|                                                                   | MCDBR       | 10.3   | 10.3  | 11.6  | 12.5  | 12.7  | 12.0  | 13.2  | 11.7  | 11.5  | 12.3  | 10.5  | 11.1  | 11.1  |
|                                                                   | MCWBR       | 8.4    | 8.7   | 9.7   | 9.2   | 8.3   | 8.3   | 8.3   | 7.7   | 8.3   | 9.8   | 9.1   | 9.6   | 9.6   |
|                                                                   | MCWBR       | 8.4    | 8.7   | 9.7   | 9.2   | 8.3   | 8.3   | 8.3   | 7.7   | 8.3   | 9.8   | 9.1   | 9.6   | 9.6   |
| Clear-Sky Solar Irradiance                                        | taub        | 0.389  | 0.289 | 0.316 | 0.346 | 0.351 | 0.340 | 0.346 | 0.335 | 0.327 | 0.291 | 0.372 | 0.390 | 0.390 |
|                                                                   | taud        | 2.544  | 2.156 | 2.261 | 2.276 | 2.364 | 2.468 | 2.492 | 2.529 | 2.476 | 2.263 | 2.630 | 2.600 | 2.600 |
|                                                                   | Ebn at Noon | 69     | 181   | 237   | 259   | 270   | 277   | 272   | 264   | 237   | 180   | 75    | 22    | 22    |
|                                                                   | Edh at Noon | 7      | 17    | 25    | 32    | 32    | 30    | 29    | 25    | 21    | 15    | 7     | 3     | 3     |
| All-Sky Solar Radiation                                           | RadAvg      | 25     | 145   | 496   | 1022  | 1443  | 1682  | 1401  | 1068  | 592   | 221   | 45    | 6     | 6     |
|                                                                   | RadStd      | 2      | 17    | 56    | 99    | 124   | 182   | 121   | 93    | 41    | 23    | 3     | 0     | 0     |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
|                         |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (11 neighbors) |  | +0.94 | +1.73   | +1.98  | N/A     | N/A   | N/A   | -218        | -266  | N/A   | N/A   |

Nomenclature: See separate page